PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

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Contents

1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4[1], the RIKEN internal technical note `Detector-PDC.pdf`[2], and the Kahlbow PhD thesis (TU Darmstadt, 2019)[3] which documents an independent reconstruction of the same hardware.

- **Design specifications.** Combining Kobayashi 2013 §3.4[1] with the SAMURAI 2012 construction proposal[4], the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$[1]; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal[5].

- **Our measured σ_y(PDC2).** In our Geant4 + reconstruction chain we observe $\sigma_y(\text{PDC2}) =$ 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration[6]. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification[1].

- **Air MS dominates and the configuration choice matters.** A quadrature decomposition[6] attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, ∼4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window → PDCs, ∼2 m). The stage D inputs from the same memo[7] give $\sigma_y(\text{PDC2}) = 7.49$ mm if Region A is vacuum and $\sigma_y(\text{PDC2}) = 13.50$ mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

The deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm, which corresponds to yoke-local coordinates $(-545.5, 0, -919.9)$ mm under the $-30°$ yoke rotation about Y[8]. Because the SAMURAI dipole cavity (`vchamber_box`) extends over $z \in [-1750, +1750]$ mm

[1] T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: 10.1016/j.nimb.2013.05.089.

[2] RIKEN Nishina Center, *Detector-PDC* technical note, `https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf`.

[3] J. Kahlbow, PhD thesis, TU Darmstadt (2019), TUprints 9192, `https://tuprints.ulb.tu-darmstadt.de/9192/`.

[4] SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), `https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf`.

[5] SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), `https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf`.

[6] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

[7] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

[8] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §1.1–§1.3 for coordinates, yoke rotation, and Region A/B definitions.

in yoke-local frame, the target at $z = -919.9$ mm is *physically inside the dipole cavity* rather than upstream of it. Two regions therefore separate the target from the PDCs. **Region A** (≈ 4.3 m) covers the in-cavity flight from the target through the cavity exit (2.67 m) and onwards through the downstream beam-pipe (`VacuumDownstream`, ≈ 1.6 m) up to the Exit Window; this region is controlled by the Geant4 macro switch `/samurai/geometry/BeamLineVacuum`. **Region B** (≈ 2.0 m) covers free flight in the world volume from the Exit Window at world $(170.5, 0, 3187.5)$ mm to PDC1 at $(700, 0, 4000)$ mm (0.97 m) and on to PDC2 (1.00 m), gated by `/samurai/geometry/FillAir`. Inside Region A the proton bends $\sim 30°$ in the 1.15 T field, but the arc-length vs. straight-line difference stays below 10%[8].

2.2 Measured σ values

Table 1 reproduces the robust half-width $\sigma = (p_{84} - p_{16})/2$ measured at PDC1 and PDC2 across three vacuum/air configurations and three truth momenta, with $|\vec{p}| = 627$ MeV/c protons.

Condition	(p_x, p_y) [MeV/c]	N	σ_x(P1)	σ_y(P1)	σ_x(P2)	σ_y(P2)	σ_{θ_x}	σ_{θ_y}
all_air	$(0, 0)$	50	3.071	9.736	3.669	12.826	12.279	9.240
all_air	$(100, 0)$	50	3.838	10.492	5.377	12.751	15.768	8.277
all_air	$(0, 20)$	50	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	$(0, 0)$	50	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	$(100, 0)$	50	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	$(0, 20)$	49	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	$(0, 0)$	50	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	$(100, 0)$	50	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	$(0, 20)$	50	0.174	0.411	0.396	1.002	6.913	2.299

Table 1: σ values from Geant4 + reconstruction ensemble; σ in mm, σ_θ in mrad. Robust half-width $(p_{84} - p_{16})/2$. Source: `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

2.3 Gap to design specifications

The realistic-air spatial σ_y(PDC2) sits roughly an order of magnitude away from the 1 mm hardware design figure, and only the all-vacuum ablation reaches the design floor. Table 2 summarises the comparison; angular and momentum entries follow the same structure.

Quantity	Design / commissioning	Our current state	Source
Position σ at PDC	1 mm	$\sim$ 13 mm (all-air) / $\sim$ 7.5 mm (A=vac) / $\sim$ 1.0 mm (all-vac)	Kobayashi 2013 §3.4[1] + memo §4[6]
Angular σ	2 mrad	5.4–9.2 mrad (σ_{θ_y}, all-air & `beamline_vacuum`) / 2.0–2.6 mrad (all-vacuum)	Kobayashi 2013 §3.4[1] + memo §4[6]
RI commissioning $\Delta p/p$	1/1500	not measured in this ablation	Kobayashi 2013[1]
Proton-mode design $\Delta p/p$	$\sim 1/700$	not measured in this ablation	RIBF-TAC05 (2005)[5]

Table 2: PDC reconstruction performance: design / commissioning targets vs. our current Geant4 + reconstruction state. Position and angular rows compare against the Kobayashi 2013 §3.4 hardware specification; the two $\Delta p/p$ rows are not yet propagated end-to-end in our ablation, so we have σ_y(PDC2) but no single $\Delta p/p$ figure to report.

2.4 Why this matters for the science floor

The PDC2 σ_y entries above are not the end of the story: they propagate backwards through the Runge–Kutta tracker and the SAMURAI magnetic field map to a target-momentum uncertainty $\sigma_{p,\mathrm{target}}$, which is the reconstruction science floor for any experiment in which the target sits inside the magnet and the outgoing proton is measured downstream (target $\to$ magnet $\to$ PDC). This document does not re-derive $\sigma_{p,\mathrm{target}}$ propagation values; the per-method numerical breakdown lives in the RK error-analysis deep-dive memo[9].

3 PDC reconstruction reference frame

3.1 Hardware foundations

The PDC1/PDC2 hardware specification is documented across four canonical sources: Kobayashi 2013 §3.4[1], the RIKEN internal `Detector-PDC.pdf` tech note[2], the SAMURAI 2012 construction proposal[4], and the older RIBF-TAC05 (2005) construction proposal[5]. The proton-arm magnet is the SAMURAI superconducting H-shape dipole described by Sato et al., providing a 7 Tm bending power, a maximum field of 3 T, an 80 cm pole gap and a 2 m pole face, with the field map computed in OPERA-3d / TOSCA[10]. The drift cells follow the Walenta geometry with an 8 mm drift distance, arranged as five planes (U/V/X/U/V) at $\pm 45°/0°$ over a 1700×800 mm^2 active area, filled with Ar + 25% iC_4H_{10} (or alternatively Ar + 50% C_2H_6) and read out via ∼810 channels[2]. The chambers sit at $\sim 59°$ on the proton arm with the design targets of 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution, with commissioning RI runs reaching $\Delta p/p \approx 1/1500$[1] and a proton-mode design specification of $\Delta p/p \approx 1/700$[5]. The signal chain feeds an ASD shaper into AMT-VME TDCs at 0.78 ns/ch[5], where the AMT chip is the TDC ASIC originally described by Arai[11]. The boundary between the dipole-cavity vacuum (Region A) and the downstream air volume (Region B) is the SAMURAI Exit Window described by Shimizu et al.: a Kevlar foil of 280 µm backed by a Mylar foil of 75 µm[12]. The matching offline reconstruction pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13]. Together these four canonical sources cover the hardware geometry and design specifications, but none of them publishes a head-to-head proton-arm reconstruction performance paper; that gap is the motivation for §3.2 (algorithm spine) and §3.3 (performance achieved).

[9] See `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pdf` for $\sigma_{p,\mathrm{target}}$ propagation under MC, Profile, Laplace, MCMC, and MC-PDC.

[10] H. Sato et al., IEEE Trans. Appl. Supercond. **23**, 4500308 (2013), DOI: 10.1109/TASC.2012.2237225.

[11] Y. Arai, Nucl. Instr. Meth. A **453**, 365 (2000), DOI: 10.1016/S0168-9002(00)00658-6.

[12] Y. Shimizu et al., Nucl. Instr. Meth. B **317**, 739–742 (2013), DOI: 10.1016/j.nimb.2013.08.059.

[13] H. Otsu et al., Nucl. Instr. Meth. B **376**, 175 (2016), DOI: 10.1016/j.nimb.2016.02.056.

Item	Value	Source
PDC location	$\sim 59°$ on proton arm	Kobayashi 2013 §3.4
Active area	1700×800 mm^2	Detector-PDC tech note
Drift cell	Walenta, 8 mm drift	Detector-PDC tech note
Plane configuration	U/V/X/U/V at $\pm 45°/0°$	Detector-PDC tech note
Primary gas	Ar + 25% iC_4H_{10}	Detector-PDC tech note
Alternative gas	Ar + 50% C_2H_6	Detector-PDC tech note
Channels	~ 810	Detector-PDC tech note
Readout TDC	AMT-VME, 0.78 ns/ch	RIBF-TAC05 2005 + Arai 2000
Magnet type	superconducting H-dipole	Sato 2013
Bending power	7 Tm	Sato 2013
Max field	3 T	Sato 2013
Gap	80 cm	Sato 2013
Pole face	2 m	Sato 2013
Field map source	OPERA-3d / TOSCA	Sato 2013
ExitWindow	Kevlar 280 µm + Mylar 75 µm	Shimizu 2013
Design position σ	1 mm	Kobayashi 2013 §3.4
Design angle σ	2 mrad	Kobayashi 2013 §3.4
Design $\Delta p/p$	0.8%	Kobayashi 2013 §3.4
Commissioning RI $\Delta p/p$	1/1500	Kobayashi 2013
Proton mode $\Delta p/p$ design	$\sim 1/700$	RIBF-TAC05 2005

Table 3: PDC1/PDC2 + magnet + ExitWindow hardware specifications. Sources cited inline; full bibliographic entries in §7.

3.2 Algorithm spine

The published reference reconstruction chain for the SAMURAI proton arm follows a four-stage spine. **(i) Hits to track segments.** Drift times measured by the AMT-VME TDCs are converted to drift distances and combined into a straight-line track in each PDC plane; this drift-time-to-distance and per-plane line-fit pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13], deployed within the SAMURAIANA / `anaroot` framework (cite by repo path `anaroot/`). **(ii) Back-tracking through the magnet.** The PDC-plane track is propagated backwards through the SAMURAI superconducting H-dipole using a Runge–Kutta integrator over the OPERA-3d / TOSCA field map published by Sato et al.[10]; the field map itself is peer-reviewed, but the back-tracking algorithmic recipe is documented only in PhD theses (see point iii) and not in any peer-reviewed paper. **(iii) Optional Kalman smoothing.** A Kalman-filter smoother on the PDC tracks is the standard refinement, but *PDC-side Kalman smoothing has no peer-reviewed publication; the full methodology lives only in PhD theses.* The closest published paper, Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018)[14], applies a Kalman filter to the MINOS TPC, not the PDCs. The PDC-side methodology is recorded across four PhD theses: Santamaria 2015 (Univ. Paris-Saclay)[15], Kahlbow 2019 (TU Darmstadt)[3], Lehr ca. 2023 (TU Darmstadt)[16], and Storck 2023 (TU Darmstadt)[17]. **(iv) Output.** The propagated state at the target plane yields the target-frame momentum vector $\vec{p}_{\text{target}}$ used downstream. As context, several MINOS+SAMURAI groups generate their training and validation data using the NPTool simulation framework of Matta et al., J. Phys. G **43**, 045113 (2016)[18], but NPTool is a simulation framework, not the reconstruction path itself.

[14] C. Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018), DOI: 10.1016/j.nima.2018.07.053.

[15] C. Santamaria, PhD thesis, Univ. Paris-Saclay (2015), HAL tel-01231191, `https://hal.science/tel-01231191`.

[16] C. Lehr, PhD thesis, TU Darmstadt (2023).

[17] S. Storck, PhD thesis, TU Darmstadt (2023), TUprints 23768, `https://tuprints.ulb.tu-darmstadt.de/23768/`.

[18] A. Matta et al., J. Phys. G **43**, 045113 (2016), DOI: 10.1088/0954-3899/43/4/045113.

Our internal pipeline lives in `libs/analysis_pdc_reco/` (see also `smsimulator5.5/CLAUDE.md` for the repo-level guide) and follows the same Runge–Kutta back-tracking spine, with multiple selectable RK fit modes; one concrete example is `--rk-fit-mode fixed-target-pdc-only` as exercised in the MS ablation memo[6] §3. An alternative neural-network backend is provided under `scripts/reconstruction/nn_target_momentum/`, and an RK geometry filter (`libs/qmd_geo_filter/`, with the matching skill `skills/smsim-rk-geometry-filter/`) prunes obviously off-acceptance trajectories before the fit. We do not re-derive the per-method numerical performance here: the full status, including MC, Profile, Laplace, MCMC, and MC-PDC variants, is reported in the RK status memo at `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_` and the deep-dive[9]. A detailed algorithm-by-algorithm comparison between the literature spine and our internal pipeline is deferred to §4.2 (Methodology gap).

3.3 Performance achieved across SAMURAI campaigns

Table 4 surveys the public SAMURAI papers that have used the PDC1/PDC2 chain and asks a single question: did the publication report a PDC-side σ_p/p figure that we can directly compare against our reconstruction? The answer for almost every entry is no — public papers report physics observables (cross sections, decay energies, momentum distributions) rather than the upstream tracker resolution. Cells are therefore explicitly marked "not reported" or "needs verification"; we deliberately do not invent numbers.

Experiment	Year	Channel	σ_p/p reported	Source flag	Citation
Commissioning RI	2012	RI ions	$\sim 1/1500$	reported	Kobayashi 2013[1]
dp elastic (190 MeV/u)	2013	d+p elastic	not reported in paper	needs verification	Sekiguchi PRC 89 (2014)[19]
dp elastic (294 MeV/u)	2017	d+p elastic	not reported in paper	needs verification	Sekiguchi PRC 96 (2017)[20]
SEASTAR-III ^{28}O	2017	(p,2p)/(p,pn)	not reported (0.2 MeV FWHM decay-E ↔ inferred $\sim 1\%$)	inferred	Kondo Nature 620 (2023)[21]
^{52}Ca(p,pn)^{51}Ca	2017	(p,pn)	not reported	needs verification	Enciu PRL 129 (2022)[22]
^{30}F study	2017	(p,2p)	not reported	needs verification	Kahlbow PRL 133 (2024)[23]
^{50}Ar(p,pn)^{49}Ar	2017	(p,pn)	not reported	needs verification	Linh PRC 109 (2024)[24]

Table 4: PDC-side σ_p/p as reported (or not) in public SAMURAI papers. Cells marked "needs verification" or "inferred" should not be filled with invented numbers. The most complete public PDC-side performance numbers are in the Kahlbow 2019 PhD (TUprints 9192)[3]. Storck 2023 PhD (TUprints 23768)[17] covers a SAMURAI December 2016 campaign with PDC analysis but with a W/Pt target rather than MINOS LH_2.

The honest reading of Table 4 is that almost every public SAMURAI paper reports physics observables (cross sections, decay energies) but not σ_p/p directly, so the "needs verification" cells are deliberate placeholders for follow-up commits and must never be filled with invented numbers. The Kondo entry is the lone borderline case: the 0.2 MeV FWHM decay-energy resolution lets one *infer* $\sigma_p/p \sim 1\%$ at the order-of-magnitude level, but that figure is inferred and is labelled as such. The recommended path to resolve the remaining cells is the Kahlbow 2019 PhD thesis[3], which contains the most complete public PDC-side performance breakdown; Storck 2023[17] is partially relevant but covers a different (W/Pt) target.

4 Gap analysis

4.1 Performance: σ values

The published reference frame sets the bar at the Kobayashi 2013 §3.4 design specification: 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution at the PDC, with the SAMURAI commissioning RI-beam runs reaching $\Delta p/p \approx 1/1500$[1]. The most recent public physics-grade comparison point is the SEASTAR-III ^{28}O measurement of Kondo et al.[25], which quotes a 0.2 MeV FWHM decay-energy resolution; propagating that to a PDC-side relative momentum resolution gives $\sigma_p/p \sim 1\%$ at the order-of-magnitude level, but the figure is inferred (not directly reported in the paper) and is flagged as such here.

In our own Geant4 + reconstruction chain, the directly observable spatial spread is $\sigma_y(\text{PDC2}) =$ 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration; the corresponding σ_{θ_y} falls in 5.4–9.2 mrad for the all-air and `beamline_vacuum` cases and 2.0–2.6 mrad for the all-vacuum case[6]. We do not introduce new numbers in this subsection; all entries are reproduced from Table 1 in §2.

The position σ_y at PDC2 is therefore an order of magnitude away from the 1 mm hardware design figure in both the all-air and the `beamline_vacuum` configurations, and the angular σ_{θ_y} is 2–4× the 2 mrad design value in those same two configurations. Only the all-vacuum ablation reaches the design floor: $\sigma_y(\text{PDC2}) \approx 1.0$ mm matches the hardware specification to within the ablation precision, and $\sigma_{\theta_y} = 2.0$–2.6 mrad straddles the 2 mrad target. We do not yet have an end-to-end σ_p/p figure from this ablation; the per-method $\sigma_{p,\text{target}}$ propagation methodology under MC, Profile, Laplace, MCMC, and MC-PDC is documented in the RK error-analysis deep-dive memo[9] and is not re-derived here.

The interpretation of this gap structure matters for how follow-up work is scoped. The fact that the all-vacuum ablation lands on the 1 mm hardware design floor is direct evidence that the reconstruction methodology itself is sound: the same RK back-tracking + drift-chamber pipeline that produces the 13 mm number in air produces the 1 mm number in vacuum, with no algorithmic change between the two. The gap to design is therefore primarily a material-budget problem rather than a methodology problem, and the responsibility for closing it sits with §4.3 (material budget and vacuum configuration). The $\sigma_y(\text{PDC2})$ entries propagate backwards through the RK tracker and the SAMURAI field map to a target-momentum uncertainty $\sigma_{p,\text{target}}$, and that propagation is captured method-by-method in the RK deep-dive[9].

4.2 Methodology

The published methodology spine for the SAMURAI proton arm has three pieces. RK back-tracking through the Sato 2013 OPERA-3d / TOSCA field map[10] forms the propagator; the front-end drift-chamber pipeline (drift-time-to-distance, per-plane line fit) is the RIKEN ROOT-based analysis of Otsu et al.[13]; an optional Kalman smoother on the PDC tracks is the standard refinement step but has *no peer-reviewed paper covering the PDC-side methodology.* The closest published Kalman paper is Santamaria et al. Nucl. Instr. Meth. A **905** (2018)[14], which applies a Kalman filter to the MINOS TPC rather than the PDCs. The actual PDC-side Kalman methodology lives only in PhD theses: Santamaria 2015[15], Kahlbow 2019[3], Lehr 2023[16], and Storck 2023[17].

Our internal pipeline lives in `libs/analysis_pdc_reco/` (with the repo-level guide in `smsimulator5.5/CLAU` and matches the literature spine piece-for-piece: it provides the same RK back-tracking propagator, exposes multiple selectable RK fit modes (one concrete example being `--rk-fit-mode fixed-target-pdc-only` as exercised in the MS ablation memo[6]), wraps a neural-network backend under `scripts/reconstruction/nn_target_momentum/`, and runs a geometry filter (`libs/qmd_geo_filte`

[25] Y. Kondo et al., Nature **620**, 965–970 (2023), DOI: 10.1038/s41586-023-06352-6.

before the fit. The full numerical status of the RK pipeline, including MC, Profile, Laplace, MCMC, and MC-PDC variants, is recorded in the RK status memo[26] and the deep-dive[9].

The methodological gap to the published reference frame is therefore small in algorithm shape: literature and our pipeline use the same RK back-tracking and the same optional Kalman smoothing. The *actionable* gap is process-noise calibration. None of the four PhD theses or the Santamaria 2018 NIM A paper[14] publishes a process-noise covariance tuned to a specific magnet + air configuration, and there is no public table that maps a given dipole + cavity + beam-pipe + downstream-air configuration to a Kalman process-noise covariance one can drop into our smoother. The quadrature decomposition in the MS ablation memo[6] §6 (with $\sigma_A \approx 11.19$ mm from Region A and $\sigma_B \approx 7.43$ mm from Region B) is exactly the per-region calibration data needed to set process-noise inputs for our specific geometry, and it does not exist in the public literature for this configuration.

The interpretation is that the published reference frame supplies the methodology shape (RK + optional Kalman), but it does not supply the configuration-specific tuning numbers, so we have to self-calibrate the process-noise inputs against our own ablation data rather than read them off a literature table. This is a closeable gap rather than a fundamental obstacle: the calibration data already exists internally[6], and feeding it into the Kalman smoother is the next concrete methodology step.

[26] See `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_zh.pdf` for the per-method RK reconstruction status.